

A proposed all-dimensional endpoint bound for the centered ball maximal operator

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Abstract

We present a proof argument for the endpoint gradient bound of the centered ball maximal operator in every fixed dimension $d \geq 2$. A real matrix correction and balanced radial mixtures provide local cancellation estimates. Positive refinement and a finite-state direction law organize these estimates across scales, while a packing bound controls spatial suprema and histories before particle birth. Cancellation of completed histories before integration by parts yields a signed finite-band estimate uniform in the number of scales. Smooth approximation and localization then pass to Sobolev inputs and remove the singular part of the limiting gradient measure.

1 Statement and notation

Fix $d \geq 2$. Norms and balls are Euclidean unless a subscript ∞ is displayed; $v_d = |B(0, 1)|$. Write

$$A_r f(x) = \frac{1}{v_d r^d} \int_{B(x, r)} f(y) \, dy, \quad Mf(x) = \sup_{r>0} A_r |f|(x).$$

All constants denoted C_d are finite and may change between lines. They must be independent of the function, the scale, and the number of scales. A subscript indicating additional fixed parameters permits dependence on those parameters.

Theorem 1.1 (Proposed endpoint conclusion). *There is $0 < C_d < \infty$ such that every real $f \in W^{1,1}(\mathbb{R}^d)$ satisfies*

$$Mf \in W_{\text{loc}}^{1,1}(\mathbb{R}^d), \quad \int_{\mathbb{R}^d} |\nabla Mf| \leq C_d \int_{\mathbb{R}^d} |\nabla f|.$$

Here $Mf \in L^{d/(d-1)}(\mathbb{R}^d)$, but global L^1 integrability of Mf is neither asserted nor used.

The proposed proof proceeds through the stronger signed finite-band estimate

$$\int |\nabla S_{a,2^m a} g| \leq C_d \int |\nabla g|, \quad S_{a,b} g = \sup_{a \leq r \leq b} A_r g, \quad g \in C_c^\infty(\mathbb{R}^d), \quad a > 0, \quad m \in \mathbb{N} \cup \{0\}, \quad (1.1)$$

then supplies the passage to a weak gradient. No assumption of (1.1) is made in proving the local or packing lemmas. Standard background used is the L^p maximal inequality for $p > 1$, the classical Sobolev inequality, cube Poincaré inequalities, smooth approximation in $W^{1,1}$, and the representation and local lower semicontinuity of distributional variation. Strong continuity of $f \mapsto \nabla Mf$ and Dini kernels are outside this draft.

2 Moments and a real matrix correction

Let g be smooth, $h = \nabla g \, dh$, $b = y - x$, and $H_r = \int_{B(x,r)} b \otimes dh$. Integration by parts gives

$$H_r^\top = H_r, \quad \partial_r A_r g(x) = \frac{\text{tr } H_r}{v_d r^{d+1}}, \quad \nabla A_r g(x) = \frac{h(B(x,r))}{v_d r^d}. \quad (2.1)$$

For the first assertion, integrate the divergence of $g(b_i e_j - b_j e_i)$: its generating vector field is tangent to the sphere and has zero divergence. The second and third assertions follow by differentiating the integral over the unit ball.

For $a \neq 0$ put

$$T_a(b) = \frac{(a \cdot b)I + ab^\top - ba^\top}{|a|^2}. \quad (2.2)$$

Then $T_a(a) = I$ and T_a is linear in b . In fact,

$$\|T_a(b)\|_{\text{op}} = \frac{|b|}{|a|}. \quad (2.3)$$

To see this, write $a = Ae$, $|e| = 1$, $b = \beta e + v$, $v \perp e$. On $\text{span}(e, v)$ the operator has matrix $A^{-1} \begin{pmatrix} \beta & |v| \\ -|v| & \beta \end{pmatrix}$; on its orthogonal complement it is multiplication by β/A . If $v = 0$ the whole operator is $\beta I/A$. This proves (2.3), including the degenerate case.

For a scalar kernel K let $H_K = \int K b \otimes dh$. Direct expansion gives

$$\int K(y) T_a(y-x) \, dh(y) = \frac{(H_K^\top - H_K)a + a \text{tr } H_K}{|a|^2}. \quad (2.4)$$

Thus symmetry and zero trace suffice; $H_K = 0$ is not required. If two kernels K_0, K_1 have these moments, then, for $z \neq x$,

$$\int (K_1 - K_0) \, dh = \int W_z(y) (K_1 - K_0)(y) \, dh(y), \quad W_z(y) = I - T_{z-x}(y-x) = T_{z-x}(z-y). \quad (2.5)$$

If their difference is supported in $D(x, r, \delta) = \{y : ||y-x| - r| \leq \delta\}$, with $1 \leq r \leq 2$, $\delta \leq 1/4$, and $z \in D(x, r, \delta)$, then $\|W_z(y)\| \leq 2\rho$ on $B(z, \rho)$ and $\|W_z(y)\| \leq 10$ on the shell. These bounds follow from (2.3).

Lemma 2.1 (Balanced radial mixtures). *If $A_r g(x) \geq A_{r-\delta} g(x), A_{r+\delta} g(x)$, $r \in [1, 2]$, and $0 < \delta \leq 1/4$, there is a nonnegative density k of integral one, supported on $[r-\delta, r+\delta]$, such that*

$$K(y) = \int k(t) \mathbf{1}_{B(x,t)}(y) \, dt$$

has symmetric, trace-zero moment matrix. Moreover, $0 \leq K \leq 1$, $\text{Lip}(K) \leq C_d/\delta$, $K = 1$ inside $B(x, r-\delta)$, and $K = 0$ outside $B(x, r+\delta)$.

Proof. Normalize t^{-d-1} separately on $[r-\delta, r]$ and $[r, r+\delta]$. By (2.1), the corresponding averages M_-, M_+ of $\text{tr } H_t$ obey $M_- \geq 0 \geq M_+$. If $M_- - M_+ > 0$, mix the densities with weights $-M_+/(M_- - M_+)$ and $M_-/(M_- - M_+)$. If both moments vanish, take the first density. Symmetry holds separately for every ball. Since $3/4 \leq t \leq 9/4$, the density is bounded by C_d/δ . The remaining conclusions follow by integrating the density between two radial distances. The formulas also give a Borel choice. $\square$

3 A shell lemma uniform over all radii

Lemma 3.1. *Let S, ν be finite positive Borel measures on $\mathbb{R}^d$, $U > 0$, $T \geq 0$, $0 < \rho \leq 1$, and $0 < \delta \leq \rho^2/1600$. There is a Borel set E with*

$$|E| \leq C_d \left[\left(\frac{\nu(\mathbb{R}^d)}{U} \right)^2 \frac{\delta}{\rho} + \frac{\nu(\mathbb{R}^d)}{U} \frac{\delta}{\rho^2} \left(1 + \frac{T}{U} \right) \right] \quad (3.1)$$

such that, simultaneously for all $x \notin E$ and $r \in [1, 2]$,

$$\begin{aligned} \nu(D(x, r, \delta)) &\geq U, \quad S(B(x, r - \delta)) \leq T \\ \implies \exists z \in D(x, r, \delta) : \quad (S + \nu)(D(x, r, \delta) \setminus B(z, \rho)) &\leq 2U. \end{aligned}$$

Proof. Put $v = 200\delta/\rho^2 \leq 1/8$. For $z \in \mathbb{R}^d$ and $\theta \in S^{d-1}$ the balls $B(z + L\theta, L)$ increase with L : membership is $|y - z|^2 < 2L(y - z) \cdot \theta$. Set $l = |x - z| \in [1/2, 3]$ and $\theta = (x - z)/l$. If $|y - z| \geq \rho$ and $||y - x| - l| \leq 2\delta$, then

$$||y - z|^2 - 2l(y - z) \cdot \theta| \leq 13\delta, \quad (y - z) \cdot \theta \geq \rho^2/12.$$

Its entry parameter $L_* = |y - z|^2/[2(y - z) \cdot \theta]$ therefore satisfies $|L_* - l| \leq 78\delta/\rho^2 < v/2$. Conversely, a point with $|y - z| \geq \rho$ already in $B(z + (l - v)\theta, l - v)$ obeys

$$|y - x|^2 < l^2 - 2v(y - z) \cdot \theta \leq l^2 - (200/3)\delta < (l - 2\delta)^2.$$

Define the Borel, nondecreasing function

$$H_{z,\theta}(L) = S(\{|y - z| \geq \rho\} \cap B(z + L\theta, L)) \quad (L > 0), \quad H_{z,\theta}(L) = 0 \quad (L \leq 0).$$

Let $\mathcal{C}(x, z)$ be the condition $H(l - v) \leq T$, $H(l + v) - H(l - v) \geq U$. Consider the radius-free Borel majorant

$$\mathcal{B}(x) = \frac{1}{U} \int_{1/2 \leq |x - z| \leq 3} \left[\frac{1}{U} \int \mathbf{1}_{\{|y - x| - |x - z| \leq 2\delta, |y - z| \geq \rho\}} d\nu(y) + \mathbf{1}_{\mathcal{C}(x, z)} \right] d\nu(z).$$

Take $E = \{\mathcal{B} \geq 1\}$. If the cap conclusion fails, for every z in the shell either the remote ν mass exceeds U or the remote S mass exceeds U . The first alternative makes the first term in brackets exceed one. The entry and earlier-interior comparisons just proved make the second alternative imply $\mathcal{C}(x, z)$, because $l - 2\delta \leq r - \delta$. Integrating over at least U of shell ν mass proves $x \in E$.

For fixed y, z separated by ρ , the first integrand restricts $x \in \overline{B}(z, 3)$ to a slab of thickness at most $13\delta/|y - z|$. This follows by subtracting squared distances; the resulting affine function has gradient length $2|y - z|$. The slab volume is at most $C_d\delta/\rho$. For the second integrand fix z, θ , and put $G = \min(H, T + U)$. On the permitted set of l , $G(l + v) - G(l - v) \geq U$. Translation and monotonicity give

$$\int_{\mathbb{R}} (G(l + v) - G(l - v)) dl \leq 2v(T + U).$$

For example, integrate on $[-L, L]$, cancel the overlap, and let $L \rightarrow \infty$. The set of permitted l has length at most $2v(1 + T/U)$. Polar coordinates with $l \leq 3$ contribute at most $3^{d-1}|S^{d-1}|$ times that length. Tonelli and $|E| \leq \int \mathcal{B}$ prove (3.1). $\square$

4 Discounted local estimates

Set $\alpha = 1/100$. Let Q be a unit cube, $\Omega = \{\text{dist}(y, Q) < 10\}$, g smooth, and $h = \nabla g \, dy$. Suppose, on Ω , that

$$h = \sum_i v_i \sigma_i + \sum_j \beta_j. \quad (4.1)$$

The sums are finite, v_i are unit vectors, σ_i are finite positive measures, and all component measures are absolutely continuous. They are defined on the whole space. Write $m_j = |\beta_j|(\mathbb{R}^d)$, and assign $s_j \in (0, 1]$. If $s_j < 1$, require zero total vector mass and support in a ball of radius $C_s s_j$. Put $S = \sum_i \sigma_i$, $A = \sum_j s_j^\alpha m_j$, and $u_r = A_r g$. Define

$$\begin{aligned} I(x) &= \{r \in [1, 2] : u_r(x) \geq u_t(x) \text{ whenever } t > 0, |t - r| \leq 1/4\}, \\ J(x) &= \{r \in [1, 2] : u_r(x) = \max_{1 \leq t \leq 2} u_t(x)\} \cup \{1, 2\}. \end{aligned}$$

Proposition 4.1. *With constants depending on d, C_s , and on κ in the first line, the following hold:*

$$\int_Q \sup_{r \in I(x)} (-e \cdot \nabla u_r(x))_+ \, dx \leq C_{d, C_s, \kappa} A \quad \text{if } |e| = 1, \, e \cdot v_i \geq \kappa > 0; \quad (4.2)$$

$$\int_Q \sup_{r \in J(x)} |\nabla u_r(x)| \, dx \leq C_{d, C_s} (A + S(\mathbb{R}^d)); \quad (4.3)$$

$$\int_Q \sup_{0 < t, t' \leq 4} |u_t(x) - u_{t'}(x)| \, dx \leq C_{d, C_s} (A + S(\mathbb{R}^d)). \quad (4.4)$$

An empty supremum of nonnegative quantities is zero.

4.1 Cancellation estimates for radial mixtures

Put $b = 2\alpha = 1/50$, $\eta = 1/10$, $\gamma = 1/5$, and $\delta_k = 2^{-k}$. Choose $\delta_* > 0$ with $\delta_* \leq 1/4$ and $\delta \leq \delta^{2\gamma}/1600$ for $\delta \leq \delta_*$. For kernels K_k with the bounds of Lemma 2.1, whether balanced or not, define

$$\nu_k = \sum_{s_j \geq \delta_k^2} |\beta_j|, \quad p_k = \int K_k \, d\beta, \quad \beta = \sum_j \beta_j, \quad D_k = D(x, r, \delta_k).$$

Zero mass and the Lipschitz bound yield

$$\nu_k(\mathbb{R}^d) \leq A \delta_k^{-b}, \quad |p_k| \leq C A \delta_k^{-\alpha}, \quad \left| \sum_{s_j < \delta_k^2} \int F \, d\beta_j \right| \leq C A \delta_k^{1-b} \quad \text{if } \text{Lip}(F) \leq C/\delta_k. \quad (4.5)$$

The test F may be scalar or matrix valued. Indeed, subtract its value at the support center. For the middle estimate use $\min(1, s/\delta) \leq (s/\delta)^\alpha$; for the last use $\sum_{s_j < \delta^2} s_j m_j \leq \delta^{2(1-\alpha)} A$. In particular, with a bounded away from zero,

$$|p_{k+1} - p_k| + \left| \int T_a(y - x)(K_{k+1} - K_k)(y) \, d\beta(y) \right| \leq C(\nu_k(D_k) + A \delta_k^{1-b}). \quad (4.6)$$

Here $K_{k+1} - K_k$ and its product with the affine matrix are globally Lipschitz, supported in $B(x, 3)$, and vanish outside D_k . For fixed x, r , dominated convergence against every absolutely continuous component gives the sharp-ball limit $K_\infty = \mathbf{1}_{B(x, r)}$. Every sphere is null for each component, so this conclusion requires no radius-dependent exceptional set.

4.2 Proof of the directional bound

If $A = 0$ the cone gives the assertion directly. Otherwise fix $\lambda \geq L_0 A$, write $q = A/\lambda$, and choose k_0 with $q/2 < \delta_{k_0} \leq q \leq \delta_*$. At each $r \in I(x)$ take balanced kernels and set $P_k = \int K_k dh$, $N_k = (-e \cdot P_k)_+$, $U_k = \lambda \delta_k^\eta$. Call k active if $\nu_k(D_k) \geq U_k$. An inactive increment in (4.6) is at most $C\lambda \delta_k^\eta$.

Apply Lemma 3.1 with ν_k , $\rho = \delta_k^\gamma$, U_k , and $T_k = C_0 \lambda \delta_k^{-b}$, where C_0 will be fixed below. The exceptional sets are independent of r and satisfy

$$|E_k| \leq C(q^2 \delta_k^{14/25} + q \delta_k^{9/25}). \quad (4.7)$$

The exponents are respectively $1 - \gamma - 2b - 2\eta$ and $1 - 2\gamma - 2b - 2\eta$; the other exponent from the second term is larger and can be replaced by $9/25$.

Fix x outside their union. The cone implies

$$e \cdot P_k \geq \kappa \int K_k dS + e \cdot p_k, \quad N_k \leq |p_k|. \quad (4.8)$$

The first active index, or ∞ if there is none, therefore has

$$N_k \leq C(A \delta_{k_0}^{-\alpha} + \lambda \delta_{k_0}^\eta). \quad (4.9)$$

Now let $k < l$ be consecutive active indices, allowing $l = \infty$ after the last active index. The basic bound at $k, k+1$ and the inactive increments between them give

$$|p_k| + |p_l| \leq C(A \delta_k^{-\alpha} + \lambda \delta_k^\eta) \leq C\lambda \delta_k^{-b}.$$

It suffices to consider $N_l > N_k$. Then $e \cdot P_l < 0$, and (4.8) bounds $\int K_l dS$ by $\kappa^{-1}|p_l|$. Since $K_l = 1$ on $B(x, r - \delta_k)$, this is at most T_k upon choosing C_0 large enough. The shell lemma gives $z \in D_k$ with

$$(S + \nu_k)(D_k \setminus B(z, \delta_k^\gamma)) \leq 2U_k. \quad (4.10)$$

If $e \cdot P_k \leq 0$, set $K_* = K_k$. Otherwise choose a convex combination of K_k, K_l with $e \cdot \int K_* dh = 0$. In both cases $N_* = N_k$, both kernels are balanced, and

$$\int (K_l + K_*) dS \leq C\lambda \delta_k^{-b}. \quad (4.11)$$

Use (2.5) with W_z . The positive component costs at most $C\lambda \delta_k^{\gamma-b} + CU_k$ by (4.10)–(4.11). For the error component, $K_l - K_*$ is a scalar in $[0, 1]$ times $K_l - K_k$. Telescope the latter through finite consecutive widths, keeping W_z fixed. At the first increment the coarse pieces cost $C\delta_k^\gamma \nu_k(\mathbb{R}^d) + CU_k$, and the fine pieces cost $CA \delta_k^{1-b}$. At each subsequent, inactive increment use (4.6) and $W_z = I - T_{z-x}(y - x)$. Their sum is bounded by $C\lambda \delta_k^\eta$. The case $l = \infty$ follows by dominated convergence, never by applying a Lipschitz estimate directly to a sharp kernel. Thus

$$N_l \leq N_k + C\lambda \delta_k^\eta, \quad \gamma - b = 9/50 > \eta.$$

Summing over active indices and using (4.9) proves

$$(-e \cdot h(B(x, r)))_+ \leq C\lambda(q^{1-\alpha} + q^\eta) < \lambda$$

after increasing L_0 . This holds for every $r \in I(x)$ at the same center. Summing (4.7) proves the tail estimate

$$|\{x \in Q : \sup_{r \in I(x)} (-e \cdot h(B(x, r)))_+ > \lambda\}| \leq C(q^{64/25} + q^{34/25}).$$

Both powers exceed one. Layer cake, the bound $|Q| = 1$ below $L_0 A$, and (2.1) prove (4.2).

4.3 Band maxima, including forced endpoints

Put every positive component into the errors with assigned size one; the price becomes $A_0 = A + S(\mathbb{R}^d)$. Repeat the preceding construction with no positive part. For $r \in J(x)$ use a balanced kernel when $\text{dist}(r, \{1, 2\}) \geq \delta_k$, and otherwise the uniform radial mixture on $[r - \delta_k, r + \delta_k]$. Exclude in addition

$$F_k = \bigcup_{a=1,2} \{x : \nu_k(\{y : ||y - x| - a| \leq 2\delta_k\}) \geq \lambda\delta_k^\eta\}.$$

The volume of each annulus of centers is at most $C_d\delta_k$, so $|F_k| \leq C(A_0/\lambda)\delta_k^{22/25}$. Outside F_k , every active width is away from the band endpoints, so both K_k, K_{k+1} are balanced. Apply the shell lemma with $S = T = 0$. At an active increment the matrix correction and (4.10) give $|P_{k+1} - P_k| \leq C\lambda\delta_k^\eta$. At an inactive increment (4.6) gives the same bound. Summation from k_0 and layer cake give (4.3); the additional tail power is $1 + 22/25 = 47/25 > 1$.

4.4 Value oscillation at small radii

Take a fixed smooth probability density ψ in the interior of Q , put $c = \int g\psi$, and $E_Q = \{\text{dist}(x, Q) < 4\}$. The fundamental theorem of calculus along segments gives

$$g(x) - c = \int L(x, y) \cdot dh(y), \quad L(x, y) = -(y - x) \int_0^1 \psi(x + (y - x)t) t^{-d-1} dt.$$

For $x \in E_Q$, this globally defined kernel is supported, as a function of y , on segments joining x to $\text{supp } \psi$, hence inside Ω . It satisfies

$$|L(x, y)| \leq C_d |x - y|^{1-d}, \quad |\nabla_y L(x, y)| \leq C_d |x - y|^{-d}.$$

Indeed nonzero integrands require $t \geq |x - y|/C_d$, and integration of t^{-d-1} and differentiation give the bounds. Choose $p = 2d/(2d - 1)$; then $1 < p < d/(d - 1)$ and $d/p - (d - 1) = 1/2$. Uniformly in y , $\|L(\cdot, y)\|_{L^p(E_Q)} \leq C_d$. For a zero-mass piece in $B(z, a)$, $a = C_s s$, use Minkowski on $B(z, 2a)$ and subtract $L(x, z)$ outside that ball. The respective bounds are

$$C_d a^{1/2} m, \quad \|C_d a m |x - z|^{-d}\|_{L^p(|x - z| > 2a)} = C_d a^{1/2} m.$$

Thus its contribution is at most $C_d C_s s^{1/2} m \leq C s^\alpha m$. Size-one pieces and the positive measures use the uniform kernel norm directly. Consequently $\|g - c\|_{L^p(E_Q)} \leq C(A + S(\mathbb{R}^d))$. All pairings are justified almost everywhere by these same estimates. For $f_Q = (g - c)\mathbf{1}_{E_Q}$ and $x \in Q$, every ball of radius at most four is in E_Q , so the oscillation is at most $2M f_Q(x)$. The strong L^p maximal inequality and Holder on Q prove (4.4).

The gradient suprema above are Borel: I, J have closed graphs with compact radius fibers, and $u_r, \nabla u_r$ are continuous. For the value supremum one can use rational radii. This completes the proof of Proposition 4.1.

5 Finite refinement and summable prices

For the rest of the smooth argument let $g \in C_c^\infty(\mathbb{R}^d)$, $h = \nabla g dx$, and $V = \int |\nabla g|$. At every integer level let $r_k = 2^{-k}$, $V_k = r_k \mathbb{Z}^d$, and

$$Q_p = p + [0, r_k]^d, \quad \phi_p(x) = \prod_{i=1}^d (1 - |x_i - p_i|/r_k)_+, \quad h_p = \int \phi_p dh = a_p \theta_p.$$

Use a fixed unit direction when $a_p = 0$. For $p \in V_i, q \in V_k, i < k$, put $\lambda_{pq} = \phi_p(q)$, and use the identity matrix at equal levels. Piecewise multilinear interpolation proves

$$\sum_p \lambda_{pq} = 1, \quad \phi_p = \sum_q \lambda_{pq} \phi_q, \quad \lambda_{ps} = \sum_q \lambda_{pq} \lambda_{qs}, \quad h_p = \sum_q \lambda_{pq} h_q. \quad (5.1)$$

Positive consecutive edges have coordinate displacement at most $r_k/2$ and nested hat supports. Each vertex has 3^d positive children. Put $A_k = \sum_{p \in V_k} a_p$; then

$$0 \leq A_k \leq A_{k+1} \leq V. \quad (5.2)$$

5.1 A dimension-dependent palette and cone

Choose $\tau = 1/(100d)$ and a finite $\tau/20$ net $\mathcal{N}$ in S^{d-1} . For each $n \in \mathcal{N}$, choose unit simplex vertices $u_1, \dots, u_d$ in $n^\perp$ with

$$\sum_i u_i = 0, \quad u_i \cdot u_j = -1/(d-1) \ (i \neq j), \quad \sum_i u_i \otimes u_i = \frac{d}{d-1} P_{n^\perp}.$$

They exist by projecting the coordinate vectors onto the sum-zero hyperplane and applying an isometry onto $n^\perp$. Set

$$e_i(n) = \frac{\sqrt{3}}{2} n + \frac{1}{2} u_i, \quad c_{n,i}(z) = \frac{2}{d\sqrt{3}} n \cdot z + \frac{2(d-1)}{d} u_i \cdot z. \quad (5.3)$$

The frame identities give $z = \sum_i e_i(n) c_{n,i}(z)$ and $\sum_i |c_{n,i}(z)| \leq C_d |z|$. Also $e_i \cdot e_j \geq 1/2$ for all i, j . Call (q, n) good if $|\theta_q - n| \leq 3\tau$. For a good pair the coefficients $c_{n,i}(h_q)$ are nonnegative: $n \cdot \theta_q \geq 1/2$, $|u_i \cdot \theta_q| \leq 3\tau$, and

$$c_{n,i}(\theta_q) \geq \frac{1}{d\sqrt{3}} - \frac{6(d-1)}{d} \tau > 0.$$

This is why the palette aperture must shrink with dimension. Finally,

$$|z| \leq C_d \left(n \cdot z + \sum_i (-e_i(n) \cdot z)_+ \right). \quad (5.4)$$

The bracket is positive for every $z \neq 0$: for $n \cdot z > 0$ this is immediate; for $n \cdot z < 0$ it is at least $(d\sqrt{3}/2 - 1)|n \cdot z|$; and for $n \cdot z = 0$, the spanning simplex supplies a negative projection unless $z = 0$. Compactness of the unit sphere and homogeneity prove the inequality. All choices of the simplex are isometric, so C_d is uniform in n .

5.2 Scalar cancellation without a terminal remainder

Let $\mathcal{C}$ be the finite indexed family of all $c_{n,i}$. For $c \in \mathcal{C}$, write $\nu^c = c(h)$ and $c_p = c(h_p)$. For $1 \leq l \leq m$ define

$$d_p^c = \begin{cases} \frac{1}{2}(\sum_{q \in V_{l+1}} \lambda_{pq} |c_q| - |c_p|), & l < m, \\ \frac{1}{2}(\int \phi_p d|\nu^c| - |c_p|), & l = m. \end{cases}$$

These numbers are nonnegative. There are constant-sign residual measures ρ_p^c of mass c_p and zero-mass measures β_p^c with variation at most $2d_p^c$, supported in $\text{supp } \phi_p$, such that

$$\phi_q \nu^c = \rho_q^c + \sum_{l=k}^m \sum_{s \in V_l} \lambda_{qs} \beta_s^c \quad (q \in V_k). \quad (5.5)$$

Here is a construction covering zero masses as well. For positive measures U, W of masses P, N , remove fractions of each with common mass $\min(P, N)$; the remainder $U - W$ after removal has constant sign. Start with $\phi_p(\nu^c)_+$ and $\phi_p(\nu^c)_-$ at level m . At the next coarser level apply the same operation to the weighted positive and negative parts of the residuals. The removed pair is β_p^c and its two masses equal d_p^c . Induction and (5.1) give (5.5). Supports are nested and absolute continuity is preserved. Column sums telescope to

$$\sum_{l=1}^m \sum_{p \in V_l} d_p^c = \frac{1}{2} \left(\int |c(\nabla g)| - \sum_{p \in V_1} |c_p| \right) \leq \frac{1}{2} \|c\| V. \quad (5.6)$$

For $Q = Q_p$ at level k , put $\mathcal{R}(Q) = \{q \in V_k : |q - p|_\infty \leq 20r_k\}$ and

$$D_Q = \sum_{c \in \mathcal{C}} \sum_{q \in \mathcal{R}(Q)} \sum_{l=k}^m 2^{-\alpha(l-k)} \sum_{s \in V_l} \lambda_{qs} d_s^c.$$

Each root occurs in 41^d neighborhoods; column sums and the geometric series yield

$$\sum_{k, Q} D_Q \leq \frac{41^d}{2(1 - 2^{-\alpha})} \sum_{c \in \mathcal{C}} \|c\| V \leq C_d V. \quad (5.7)$$

Let I_k, J_k be the scaled radius sets in Proposition 4.1, and $v_k(x) = \sup_{0 < t, t' \leq 4r_k} |u_t(x) - u_{t'}(x)|$. That proposition gives

$$\int_Q \sup_{t \in I_k(x)} \sum_i (-e_i(n) \cdot \nabla u_t(x))_+ \leq C_d \left(D_Q + \sum_{q \in \mathcal{R}(Q): (q, n) \text{ bad}} a_q \right), \quad (5.8)$$

$$\int_Q \left(\sup_{t \in J_k(x)} |\nabla u_t(x)| + r_k^{-1} v_k(x) \right) \leq C_d \left(D_Q + \sum_{q \in \mathcal{R}(Q)} a_q \right). \quad (5.9)$$

To verify the application, on the $10r_k$ neighborhood of Q , $h = \sum_{q \in \mathcal{R}(Q)} \phi_q h$. Insert (5.5) in the d coordinates of (5.3). Good residuals are positive in directions whose scalar products with every testing $e_i(n)$ are at least $1/2$. Bad residuals cost $C_d a_q$ and are assigned size one. A pair at level l has support radius $\sqrt{d} r_l$ and variation at most $2\lambda_{qs} d_s^c$; assign it relative size r_l/r_k . For the size and value bounds put all residuals among size-one errors. Under $y = p + r_k X$ the gradient measure transforms by $r_k^{1-d} (T^{-1})_\# h$; the gradient integral has that same factor, while the value integral has factor r_k^{-d} . These identities prove the displayed factors, including $r_k^{-1} v_k$ in (5.9).

6 Packing switches and spatial derivatives

This section uses only a compactly supported finite vector measure and the refinement identities, not the fact that it is a gradient.

6.1 Neighboring directions

For $z \in V_k$ select $w(z)$ of largest a_w among $|w - z|_\infty \leq 300r_k$, with fixed ties, and set $\Theta_z = \theta_{w(z)}$. Select $n_z \in \mathcal{N}$ with $|n_z - \Theta_z| \leq \tau/20$. Define

$$E_k(q) = \max_{z \in V_k: |z - q|_\infty \leq 200r_k} |\Theta_z - \theta_q|^2 \leq 4.$$

We claim

$$\sum_{k=1}^m \sum_q a_q E_k(q) \leq C_d V. \quad (6.1)$$

For $a_q > 0$, select a maximizing z and $w = w(z)$; then $a_w \geq a_q$ and $|w - q|_\infty \leq 500r_k$. Take $L = 12$ and a nearest coarse vertex $p \in V_{k-L}$. The interpolation coefficients of q, w are at least $(1/2)^d, (3/8)^d$ respectively. The defect

$$H_{p,k} = \sum_s \lambda_{ps} a_s - a_p = \frac{1}{2} \sum_s \lambda_{ps} a_s |\theta_s - \theta_p|^2$$

therefore bounds $a_q E_k(q)$ up to C_d . A coarse vertex is assigned only a dimension-dependent finite number of fine vertices. Sum over p to get $C_d(A_k - A_{k-L})$, then sum over k : each nonnegative increment of A_j occurs at most L times. This proves (6.1), including zero masses.

6.2 The probability rows

At x , sample $z \in V_1$ with probabilities $\phi_z(x)$ and set $n_1 = n_z$. At step $k \geq 2$, given label n , sample $z \in V_k$ with probabilities $\phi_z(x)$ and switch to n_z with probability

$$p_k(n, z) = P(|n - \Theta_z|), \quad P(t) = \min(1, \tau^{-2}(t - \tau)_+^2).$$

Otherwise keep n . A positive switching probability always changes the label. The initial and transition rows are

$$\pi_1(x) = \sum_z \phi_z(x) \delta_{n_z}, \quad \pi_k(x; n, \cdot) = \delta_n + \sum_z \phi_z(x) p_k(n, z) (\delta_{n_z} - \delta_n).$$

They are Lipschitz stochastic rows. Let $\zeta_k(x)$ be the marginal law of n_k , and put

$$\begin{aligned} \rho_k(x | n) &= \sum_z \phi_z(x) p_k(n, z), \\ R_k(x) &= \sum_n \zeta_{k-1}(x; n) \rho_k(x | n) \quad (2 \leq k \leq m), \\ T_k(x) &= r_k \sum_n \zeta_{k-1}(x; n) \sum_{n'} |\nabla_x \pi_k(x; n, n')|. \end{aligned}$$

Use the single initial row for T_1 , and $R_1 = R_{m+1} = 1$. Set derivative costs to zero on the null union of grid hyperplanes at these finitely many scales. Then $0 \leq R_k \leq 1$, $0 \leq T_k \leq C_d$, since at most 2^d hats occur off those hyperplanes and their gradients have norm at most $\sqrt{d}/r_k$.

Put $c_* = (\tau/10)^2$. If $E_k(q) \leq c_*$ and $|x - q|_\infty \leq 50r_k$, every sampled direction is within $\tau/10$ of θ_q . A newly selected label is within $3\tau/20$, and a retained label within $2\tau + \tau/10 < 3\tau$. Thus (q, n_k) is good with probability one. This assertion holds on the entire stated neighborhood.

Proposition 6.1 (Packing with cell suprema). *For $D(q) = \{x : |x - q|_\infty \leq 25r_l\}$, $q \in V_l$, one has*

$$\sum_{l=1}^m \sum_{q \in V_l} a_q \left[\sup_{D(q)} (R_l + R_{l+1}) + \sum_{k=1}^l 2^{k-l} \operatorname{ess\,sup}_{D(q)} T_k \right] \leq C_d V. \quad (6.2)$$

6.3 Particles and stopping by direction change

Unfold the finite refinement forest from the roots whose supports meet $\operatorname{supp} h$ to level m . A path prefix $\xi = (q_1, \dots, q_k)$ has weight $b_\xi = \prod_{s < k} \lambda_{q_s q_{s+1}}$ and mass $M_\xi = b_\xi a_{q_k}$. Its vector is

$b_\xi h_{q_k}$, so $M_\xi \leq \sum_{\eta \text{ child}} M_\eta$. There is a positive measure μ on full paths and birth levels B satisfying

$$\mu\{B \leq k, \text{ prefix } \xi\} = M_\xi, \quad \mu\{B \leq k, q_k = q\} = a_q, \quad \mu(\text{all}) = A_m \leq V. \quad (6.3)$$

Explicitly, at each root create its mass. If the sum of child capacities is $S > 0$, send the fraction M_η/S of every existing population type to child η and create the deficit $M_\eta - M_\xi M_\eta/S$ there. If $S = 0$, all masses vanish. Give each new particle its entire ancestral prefix, even before birth. This finite recursion proves (6.3) by mass conservation. For its terminal position $y = q_m$,

$$|y - q_s|_\infty \leq \sum_{j=s}^{m-1} r_j/2 < r_s. \quad (6.4)$$

Mark the roots, then the first descendants whose direction is more than $\tau/8$ from the last marked direction, and repeat. For a marked nonterminal ξ , let $\mathcal{L}_\xi$ be the frontier of first subsequent marks, completed by terminal nodes. Vector additivity gives

$$\sum_{\eta \in \mathcal{L}_\xi} M_\eta - M_\xi = \frac{1}{2} \sum_{\eta \in \mathcal{L}_\xi} M_\eta |\theta_\eta - \theta_\xi|^2 \geq \frac{\tau^2}{128} \sum_{\eta \in \mathcal{L}_\xi \text{ marked}} M_\eta.$$

Telescoping these frontiers and adding the root masses shows $\sum_{\xi \text{ marked}} M_\xi \leq C_d V$. Occupancy and (6.1) also give

$$\int \sum_{s=B}^m E_s(q_s) d\mu \leq C_d V. \quad (6.5)$$

Run the label law at the fixed point y of a particle. Declare a level $s \geq B$ exceptional if marked or if $E_s(q_s) > c_*$. On each consecutive remaining interval all path directions are within $\tau/8$ of a common last marked direction, even if that mark predates birth. By (6.4), all sampled directions are within $\tau/10$ of the path direction. Any two sampled directions on that interval are therefore within $9\tau/20$. After a switch, the new label is within a further $\tau/20$, so every later switching probability on the interval is zero. There is at most one switch per such interval, regardless of the entering label. With J exceptional levels the count is at most $2J + 1$. Taking expectation and integrating yields

$$\int \sum_{s=B}^m R_s(y) d\mu \leq C_d V. \quad (6.6)$$

6.4 From one point to a whole cell

Fix a particle, l , and $x \in D(q_l)$. Put $c_s = R_s(y) + E_s(q_s)$, so $0 \leq c_s \leq 5$ and $c_1 \geq 1$. For $s \leq l + 1$, every tent active at x or y , including a tent differentiated off the grid, has vertex within $54r_s$ of q_s . Their selected directions are all within $\sqrt{E_s(q_s)}$ of θ_{q_s} . The function $\sqrt{P(t)}$ is $1/\tau$ -Lipschitz. For each old label, comparison with a tent sampled at y therefore gives, after averaging at y ,

$$\begin{aligned} \rho_s(x | n) + r_s \sum_{n'} |\nabla \pi_s(x; n, n')| &\leq C_d (\rho_s(y | n) + E_s(q_s)), \\ \|\pi_s(x; n, \cdot) - \pi_s(y; n, \cdot)\|_1 &\leq C_d 2^{s-l} (\rho_s(y | n) + E_s(q_s)). \end{aligned}$$

For detail, $p_s(n, z) \leq 2p_s(n, z') + 8\tau^{-2}E_s(q_s)$; each row is a sum of $\phi_z p_s(n, z)(\delta_{n_z} - \delta_n)$, and the hats have differences at most $C_d|x - y|/r_s$ and gradients at most C_d/r_s . Only at most 2^{d+1} hats are involved in the differences. The initial row has bounds C_d and $C_d 2^{1-l}$, covered by c_1 .

Stochastic matrices contract the ℓ^1 norm of signed row vectors. Thus $d_j = \|\zeta_j(x) - \zeta_j(y)\|_1$ obeys

$$d_j \leq d_{j-1} + C_d 2^{j-l} c_j, \quad d_j \leq C_d \sum_{s \leq j} 2^{s-l} c_s.$$

First averaging row costs against $\zeta_{k-1}(y)$, then replacing this marginal by $\zeta_{k-1}(x)$, costs at most $C_d d_{k-1}$. It follows uniformly on the cell that

$$R_k(x) + T_k(x) \leq C_d \sum_{s=1}^k 2^{s-k} c_s, \quad k \leq \min(l+1, m). \quad (6.7)$$

The derivative assertion holds almost everywhere, exactly as required for essential suprema. The switch assertion holds everywhere.

6.5 Summation, including pre-birth histories

Let $\mathcal{C}_l(q)$ be the bracket in (6.2). Occupancy rewrites its left side exactly as $\int \sum_{l=B}^m \mathcal{C}_l(q_l) d\mu$. The uniform bound (6.7), the final sentinel, and

$$\sum_{k=1}^l 2^{k-l} \sum_{s=1}^k 2^{s-k} c_s = \sum_{s=1}^l (l-s+1) 2^{s-l} c_s$$

bound the inner sum by

$$C_d \left[1 + \sum_{l=B}^m \sum_{s=1}^{\min(l+1, m)} (l-s+2) 2^{s-l} c_s \right].$$

For $s \geq B$, its coefficient is at most $\sum_{j=-1}^{\infty} (j+2) 2^{-j} = 8$; integrate using (6.5) and (6.6). For $s < B$, use $c_s \leq 5$. With $q = B - s \geq 1$, $j = l - B \geq 0$, the whole pre-birth contribution for one particle is at most

$$5 \sum_{q \geq 1, j \geq 0} (j+q+2) 2^{-j-q} = 5 \sum_{n \geq 1} n(n+2) 2^{-n} = 50.$$

Its integral and the sentinel cost are bounded by $C_d \mu(\text{all})$. This proves Proposition 6.1.

7 Assembly of the signed finite band

For $1 \leq i \leq j \leq m$ put

$$F_{ij}(x) = \max_{r_j \leq t \leq 2r_i} u_t(x).$$

These are compactly supported Lipschitz functions. At a point where all the finitely many envelopes are differentiable, every attaining radius t satisfies $\nabla F_{ij}(x) = \nabla u_t(x)$: the smooth function u_t touches the differentiable envelope from below. No derivative of a chosen radius is taken.

A label history decomposes $1, \dots, m$ into maximal constant runs $[i, j]$. Their radius intervals cover $[r_m, 1]$. At least one run attains F_{1m} ; apply (5.4) to that run and add the nonnegative brackets from the others. Averaging gives

$$|\nabla F_{1m}| \leq C_d \mathbb{E}_x \sum_{[i,j] \text{ run}} \left[n_i \cdot \nabla F_{ij} + \sum_a (-e_a(n_i) \cdot \nabla F_{ij})_+ \right]. \quad (7.1)$$

7.1 Unfavorable components and endpoint terms

At an attaining radius in a bin $[r_k, 2r_k]$, an interior bin of a run has radius in $I_k(x)$; an end bin has radius in $J_k(x)$. The probability that k is an end bin is at most $R_k + R_{k+1}$. Consequently the unfavorable terms are bounded by

$$\sum_{k,n} \zeta_k(x;n) \sup_{t \in I_k(x)} \sum_a (-e_a(n) \cdot \nabla u_t(x))_+ + d \sum_k (R_k + R_{k+1}) \sup_{t \in J_k(x)} |\nabla u_t(x)|.$$

Integrate on the cells and use (5.8)–(5.9). The prices sum by (5.7). For a bad root with $E_k(q) \leq c_*$ the corresponding label probability vanishes throughout the cell, by the good-label property. Other bad roots cost at most $c_*^{-1} a_q E_k(q)$ and sum by (6.1). For end bins use $Q \subset D(q)$ and (6.2). All cell multiplicities and the palette are finite, depending only on d . The integrated unfavorable contribution is at most $C_d V$.

The same reasoning bounds the baselines at run starts:

$$\int \mathbb{E}_x \sum_{[i,j] \text{ run}} |\nabla u_{r_i}| = \sum_i \int R_i |\nabla u_{r_i}| \leq C_d V, \quad (7.2)$$

because r_i is a forced endpoint in J_i .

7.2 Subtract the old history before taking absolute values

For a fixed history $\mathbf{n}$ put

$$\Phi_{\mathbf{n}} = \sum_{[i,j] \text{ run}} n_i (F_{ij} - u_{r_i}), \quad w_{\mathbf{n}}(x) = \prod_{k=1}^m \pi_k(x; n_{k-1}, n_k),$$

using an auxiliary initial state. The vector field $\Phi_{\mathbf{n}}$ is compactly supported and Lipschitz. The weak product rule gives

$$\sum_{\mathbf{n}} \int w_{\mathbf{n}} \operatorname{div} \Phi_{\mathbf{n}} = - \sum_{\mathbf{n}} \int \Phi_{\mathbf{n}} \cdot \nabla w_{\mathbf{n}}. \quad (7.3)$$

For the step- k term in the derivative, fix a prefix $\mathbf{a} = (n_1, \dots, n_{k-1})$. Let $\Psi_{\mathbf{a}}$ be its potential with the last run closed at $k-1$, or zero for the empty prefix. Summing future products gives one, while $\sum_{n_k} \nabla \pi_k(x; n_{k-1}, n_k) = 0$. Therefore the step- k pairing in (7.3) is unchanged by replacing $\Phi_{\mathbf{n}}$ with $\Phi_{\mathbf{n}} - \Psi_{\mathbf{a}}$.

For every continuation of the prefix,

$$|\Phi_{\mathbf{n}} - \Psi_{\mathbf{a}}| \leq \sum_{l=k}^m v_l. \quad (7.4)$$

Completed runs cancel. If the last prefix run continues from $[i, k-1]$ to $[i, j]$, its baseline cancels and its envelope increases by at most $\sup_{0 \leq t \leq r_{k-1}} (u_t - u_{r_{k-1}})_+ \leq v_k$. Each new run starting at l contributes at most v_l ; the starting indices are distinct and are greater than k if the old run continued. This proves (7.4).

Now take absolute values. Prefix probabilities sum to marginal probabilities, and the future products sum to one, so

$$\left| \sum_{\mathbf{n}} \int w_{\mathbf{n}} \operatorname{div} \Phi_{\mathbf{n}} \right| \leq \sum_{1 \leq k \leq l \leq m} \int \frac{T_k(x)}{r_k} v_l(x) dx. \quad (7.5)$$

On a level- l cell use (5.9) and take the essential supremum of T_k . The price terms cost at most $C_d \sum D_Q$, because $T_k \leq C_d$ and $\sum_{k \leq l} r_l / r_k \leq 2$. The root terms are bounded by the derivative

part of (6.2), since $Q \subset D(q)$ and $r_l/r_k = 2^{k-l}$. Thus (7.5) is at most $C_d V$. Combine this with (7.2), the unfavorable estimates, and (7.1) to obtain

$$\int |\nabla F_{1m}| \leq C_d V.$$

For a band $[a, b]$ with $b/a = 2^m$, apply this to $g_b(x) = g(bx)$. Both gradient integrals scale by b^{1-d} , so (1.1) follows. For $m = 0$, $\nabla A_a g = A_a \nabla g$ and Tonelli give the bound with constant one.

8 Passage to $W^{1,1}$ and removal of singular variation

For $g \in W^{1,1}$, the translation inequality yields

$$\text{Lip}(S_{a,b}g) \leq \frac{\|\nabla g\|_1}{v_d a^d}, \quad \|S_{a,b}g - S_{a,b}\tilde{g}\|_\infty \leq \frac{\|g - \tilde{g}\|_1}{v_d a^d}. \quad (8.1)$$

Approximate g in $W^{1,1}$ by compactly supported smooth functions. Uniform convergence in (8.1) and local lower semicontinuity of variation pass (1.1) to g on every fixed dyadic band. The limiting envelope is Lipschitz, so its variation is the integral of its classical gradient.

Now take $g = |f|$, so $g \in W^{1,1}$ and $\|\nabla g\|_1 \leq \|\nabla f\|_1$. Sobolev embedding and the strong maximal inequality imply $Mg \in L^{d/(d-1)}$, hence L^1_{loc} . The functions $G_N = S_{2^{-N}, 2^N} g$ increase to Mg and converge locally in L^1 . The uniform variation bound gives a finite vector Radon measure $H = D(Mg)$ with

$$|H|(\mathbb{R}^d) \leq C_d \|\nabla g\|_1, \quad |H|(O) \leq \liminf_N \int_O |\nabla G_N| \quad (O \text{ open}). \quad (8.2)$$

This step alone gives BV regularity, not Sobolev regularity.

Let K be a compact Lebesgue-null set, fix dyadic $t = 2^{-L}$, and choose an open $K \subset O \subset \{\text{dist}(x, K) < t\}$. For $N \geq L$,

$$G_N = \max(S_{2^{-N}, t} g, S_{t, 2^N} g), \quad \int_O |\nabla S_{t, 2^N} g| \leq \frac{|O|}{v_d t^d} \|\nabla g\|_1.$$

The weak gradient rule for a maximum applies also on its equality set. To bound the small-radius term, use side- t cubes meeting O . With concentric dilations, choose $\chi_Q = 1$ on $3Q$, supported in $5Q$, and $|\nabla \chi_Q| \leq C_d/t$. Set $c_Q = |5Q|^{-1} \int_{5Q} g$, $g_Q = \chi_Q(g - c_Q)$. Cube Poincare gives

$$\|\nabla g_Q\|_1 \leq C_d \int_{5Q} |\nabla g|.$$

All radius- $\leq t$ balls centered in Q lie in $3Q$, so their signed band envelopes for g_Q and g differ by c_Q on Q . Apply the signed estimate, then bounded overlap, to get

$$\int_O |\nabla S_{2^{-N}, t} g| \leq C_d \int_{\{\text{dist}(x, K) < 8\sqrt{d}t\}} |\nabla g|.$$

By (8.2),

$$|H|(K) \leq |H|(O) \leq C_d \int_{\{\text{dist}(x, K) < 8\sqrt{d}t\}} |\nabla g| + C_d t^{-d} |O| \|\nabla g\|_1.$$

First shrink $|O|$ to zero for fixed t , then let $t \downarrow 0$. The first integral tends to zero since K is null. Thus $|H|(K) = 0$. Inner regularity and containment in Borel null sets give $H \ll dx$. Radon-Nikodym supplies $H = G dx$ with $\int |G| = |H|(\mathbb{R}^d) \leq C_d \|\nabla f\|_1$. The distributional identity makes G the weak gradient of Mf , which proves the proposed conclusion of Theorem 1.1.